

Deliverable #1 Template : Software Requirement Specification (SRS)

SE 3A04: Software Design II – Large System Design

Tutorial Number: T03

Group Number: G1

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IMPORTANT NOTES

- Be sure to include all sections of the template in your document regardless whether you have something to write for each or not
 - If you do not have anything to write in a section, indicate this by the *N/A*, *void*, *none*, etc.
- Uniquely number each of your requirements for easy identification and cross-referencing
- Highlight terms that are defined in Section 1.3 (**Definitions, Acronyms, and Abbreviations**) with **bold**, *italic* or underline
- For Deliverable 1, please highlight, in some fashion, all (you may have more than one) creative and innovative features. Your creative and innovative features will generally be described in Section 2.2 (**Product Functions**), but it will depend on the type of creative or innovative features you are including

1 Introduction

This document will provide visibility and documentation for **SCEMAS**, an environmental monitoring and alert system. This document will discuss the purpose of the system, the scope of the application, user characteristics, product requirements, and use case diagrams.

1.1 Purpose

The document focuses on software requirements, user characteristics, and use cases for **SCEMAS**.

This document is intended for internal **SCEMAS** stakeholders, including, project managers, developers, domain experts, and **SCEMAS** team members/investors. No prior readings are required.

1.2 Scope

SCEMAS, the sensor data application will allow users to view all types of sensor data for geographic regions in their city. We are developing a “**Sensor Processor**” system.

City operators are required to create an account to access the sensor service. When a **city operator** enters the application they will have the option to view all the different types of data, manage sensors, view values for specific geographic regions bounded by **geofences**, and **view hotspots**. These values are all collected and updated from different sensors through the “**Sensor Processor**” system. The **city operator** can choose to release alerts automatically created by the system. **Subscribers can use the Alert API to receive alerts for a specific geographic area and choose the sensor data type. Any actions made by a city operator will be logged through the “Audit Management” system.**

The objective of the software is to collect sensor data for geographical regions by grouping sensor data together, to provide that data for city operators, government officials, and citizens, and to release alerts whenever any type of pollution level is at a dangerous or surprisingly high level.

The goals of the **SCEMAS** application are to gather better sensor data about geographical regions, create high quality independent/region sensor data to allow for better conclusions to be drawn, and to encourage public awareness of the different types of pollution.

1.3 Definitions, Acronyms, and Abbreviations

- **SCEMAS**: Smart City Environmental Monitoring and Alert System
- **PIPEDA**: Personal Information Protection and Electronic Documents Act
- **Geofence**: A geographic area bounded by a set of coordinates.
- **TLS**: Transport Layer Security (this is a cryptographic protocol).
- **WAF**: Web application firewall; A service that sits between a web server and the wider internet that scans and sanitizes incoming HTTP requests against common exploitation patterns before forwarding them to the server.
- **TCP**: Transmission Control Protocol.
- **ICP**: Internet Computer Protocol

1.4 References

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- [2] “Communicating risk in public health emergencies,” World Health Organization, 2017. Available: <https://iris.who.int/server/api/core/bitstreams/27ef4bb6-61ab-4870-b609-c09b2710633a/content>. [Accessed: Feb. 13, 2026]
- [3] “SLA & Uptime calculator: How much downtime corresponds to 99.9 % uptime,” *uptime.is*. Available: <https://uptime.is/>. [Accessed: Feb. 13, 2026]
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- [9] Office of the Privacy Commissioner of Canada, “PIPEDA Fair Information Principles,” *Priv.gc.ca*, 2019. Available: https://www.priv.gc.ca/en/privacy-topics/privacy-laws-in-canada/the-personal-information-protection-and-electronic-documents-act-pipeda/p_principle/. [Accessed: Feb. 14, 2026]
- [10] ISO: Global standards for trusted goods and services, “ISO/IEC 20922:2016,” *ISO.org*, 2016. Available: <https://www.iso.org/standard/69466.html> [Accessed: Feb. 14, 2026]

1.5 Overview

Section 2 discusses the overall product description, this includes the product perspective, product functions, user characteristics, constraints, assumptions and dependencies, and apportioning of requirements. Section 3 contains the use case diagram for the “Define New Alert Rule” use case scenario. Section 4 discusses the highlights of functional requirements and contains all the business events and the different viewpoints relating to each business event. Section 5 discusses all the non-functional requirements. Section 6 discusses the innovative feature our team decided to implement.

2 Overall Product Description

2.1 Product Perspective

SCEMAS is a scalable software platform that sends alerts to users and third party systems of recent environmental concern in a specific region. From the large scale - rapid development of manufacturing and urban planning; environmental, ecological, and health &

safety has become an issue that is now imposed by recent development tradeoffs. The **SCEMAS** software solves this issue by continuously monitoring the environmental vitals of a region and comparing it to thresholds that can become hazardous if not managed. **SCEMAS** uses MQTT clients, i.e sensors, to take in key environmental indicators (e.g air quality, noise levels, temperature, humidity, other indicators) through **TCP**. The indicators are sent to the MQTT broker that gets authenticated and authorized, then filtered and published to the time series database. The data values get aggregated to 5-minutes averages and hourly maximums in a region and compares it to the alert rule (that contains a threshold for specific regions of specific environmental indicators). The alert rule can be added, changed, and removed by health and city administrators.

The product also allows public users to create accounts and subscribe to specific alerts, edit and also remove from the system. External systems such as traffic control and/or emergency response can also subscribe to the system. City operators have special administrative access to real-time visualization that includes region, sensor, metrics, alert, and historical data records. Regions can also be added, edited, and removed for the alert channel. Administrative roles are assigned to manage and/or observe data of the system.

Public REST API sends aggregated data to public and third-party use, whilst notification API (webhook) sends alerts when data reaches over the threshold value. Once the alert occurs, users are able to view real-time data with subscribed regions of alerts using external google map API. These alerts contain warnings and call-to-action of the subscribed region. All system processes get audited to the database for keeping historical logs of the system.

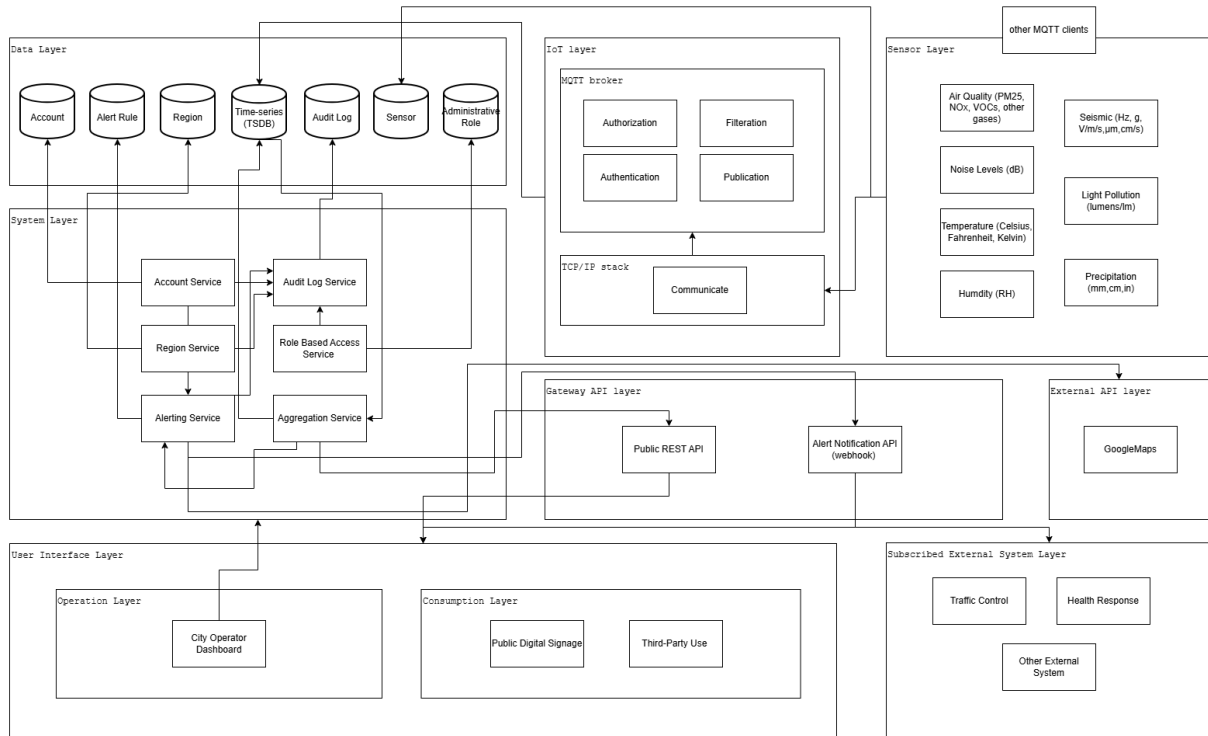


Figure 1. System Diagram

2.2 Product Functions

There will be four main modules in the product: the Dashboard Service, Sensors Registry, Alert Engine, and Account Service system. For the Dashboard Service, the primary functionality includes reading live sensor data, displaying aggregated historical data, and enabling dashboard customization for municipal government branding. Within the Sensors Registry, administrators can define geographic regions to collect sensor data for, create and register new sensors to a pre-existing region, and delete inactive sensors. Next, the Alert Engine provides administrators access to modify existing alert rules for a region, or define new alert rules within the engine. Finally, the Account Service module enables users to register with an official **SCEMAS** account, and authenticates the user upon login.

Modules	Functions
Dashboard Service	● Read live data - Allows City Operators to read live sensor data - Allows City Operators to preview system health ● Read historic data - Allows City Operators, users, and third-party developers to read historic sensor data ● Request to customize dashboard - Allows municipal governments to add custom branding to the dashboard
Sensors Registry	● Define geographic region - Allows administrators to define geographic regions to aggregate sensor data for ● Create/register sensor - Allows administrators to create and register sensors ● Delete sensor - Allows administrators to delete existing sensors
Alert Engine	● Define new alert rule - Allows administrators to set up new alert rules in the alert engine for their region ● Delete existing alert rule - Allows administrators to delete existing alert rules in the alert engine for their region
Account Service	● User register - Allows the user to create an account ● User login - Allows the user to login to the SCEMAS platform

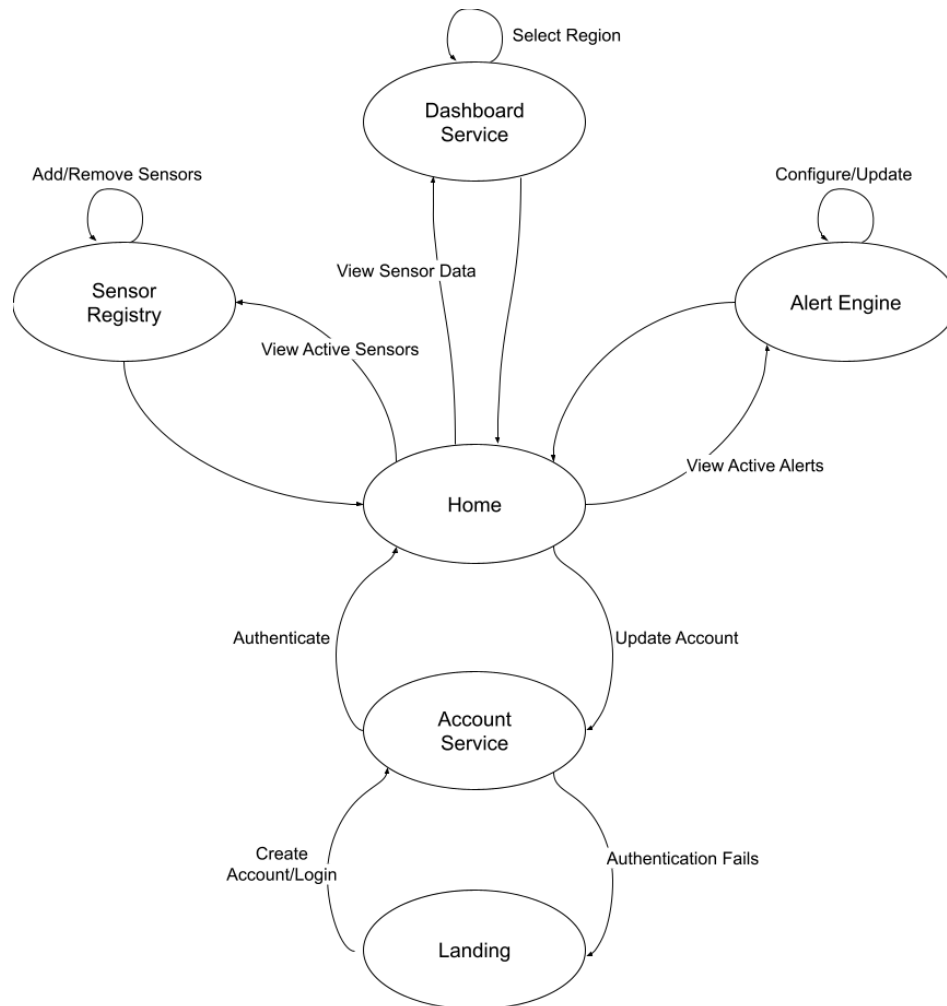


Figure 2. State Diagram

2.3 User Characteristics

SCEMAS is broadly expected to have three classes of users, operators from the city, third-party developers interacting with the API, and regular citizens interacting with third-party applications. The characteristics of these user classes are outlined below:

2.3.1 City Operators

The dashboard component of **SCEMAS** is specifically designed for use by trained city operators, and so has the following expected qualifications

1. Education Level: Basic literacy skills
 - a. Consistent with federal accessibility standards [1], the **SCEMAS** dashboard will present information in a way that can be understood by individuals with an 8th grade reading level

2. Experience: Domain-specific training
 - a. Operators working with **SCEMAS** are expected to undergo domain-specific training prior to using the platform. **SCEMAS** should strive to be easy to navigate, but should not simplify its interface to the point of interfering with an operator's workflow.
3. Technical expertise: Familiarity with desktop computers
 - a. Operators are expected to access **SCEMAS** via a desktop operating system, and should be familiar with such an interface to use the platform effectively.

2.3.2 External Developers

Data collected by **SCEMAS** will also be made available to third-party developers through an API, users accessing the system through this interface are expected to have the following qualifications:

4. Education Level: Basic literacy skills
 - a. Consistent with federal accessibility standards [1], API documentation will be written in a way that can be understood by individuals with an 8th grade reading level.
5. Experience: Familiarity with REST APIs, no domain-specific training with **SCEMAS**
 - a. Documentation of the REST API will assume prior familiarity with such APIs by its readers.
 - b. Third-party developers are not expected to undergo domain-specific training for interacting with **SCEMAS**. The interface should provide straightforward means to register a developer account and create API credentials.
6. Technical expertise: Technically literate
 - a. External developers are expected to be technically literate. Resources provided by **SCEMAS** will be limited to a straightforward way of registering and obtaining API credentials, and REST API documentation in a standard format [ex openapi]. Implementation details for interfacing with our API is left to the developer themselves to implement.

2.3.3 Private Citizen

Private citizens will be able to access data aggregated by **SCEMAS** using third-party applications, powered by our external API. While these users don't directly access our system, we still think it important to include their perspectives. The qualifications we expect from these users are:

7. Education Level: Any

- a. We expect that a variety of external applications can be developed, and data will be presented such that it can be understood by any individual.
- 8. Experience: Any
 - a. Public-facing external applications should not require any specific experience by users.
- 9. Technical expertise: Basic experience with personal technology
 - a. Public-facing applications will be available for desktop and mobile platforms, and will not require any specific technical expertise on the part of users

2.4 Constraints

1. MQTT has been prescribed for use between the **SCEMAS** system and IoT sensors
2. The system should assume operators have access to average-speed desktop machines; must be responsive enough for operators to acknowledge alerts on such machines within 30 seconds.

2.5 Assumptions and Dependencies

1. We assume **SCEMAS** will not be a single deployment, but will be distributed to customers, with them making the final deployment decision (cloud provider, self-hosted, etc). As such, we assume that per-instance customization should be possible by the customer.
2. We assume the **SCEMAS** dashboard will be accessed from modern web browsers (Chromium or firefox-based, no IE)
3. We assume that the **SCEMAS** dashboard will have a stable internet connection whenever it is being used.
4. Innovative feature: City operators should be able to create custom dashboards from standard visualization components.
5. Assume that the initial market for **SCEMAS** will be Southern Ontario
6. Assume this is a government project (this creates new requirements for the look and appearance of the application as well as the corporate regulations regarding it)
7. We are assuming these are the data types we need to enter
 - a. Air quality indicators (PM25, NOx, VOCs,)
 - b. Seismic Sensors
 - c. Noise level (dB)
 - d. Temperature (deg C)
 - e. Humidity (%RH)
 - f. Precipitation
 - g. Light pollution (lumens/lm)

2.6 Apportioning of Requirements

1. Internationalization: The initial version of **SCEMAS** will only be available in English.
2. Privacy Requirements: **SCEMAS** will initially comply with **PIPEDA**. The system may be extended later to comply with other global privacy regulations (e.g. GDPR)

3.1 Use Case Diagram

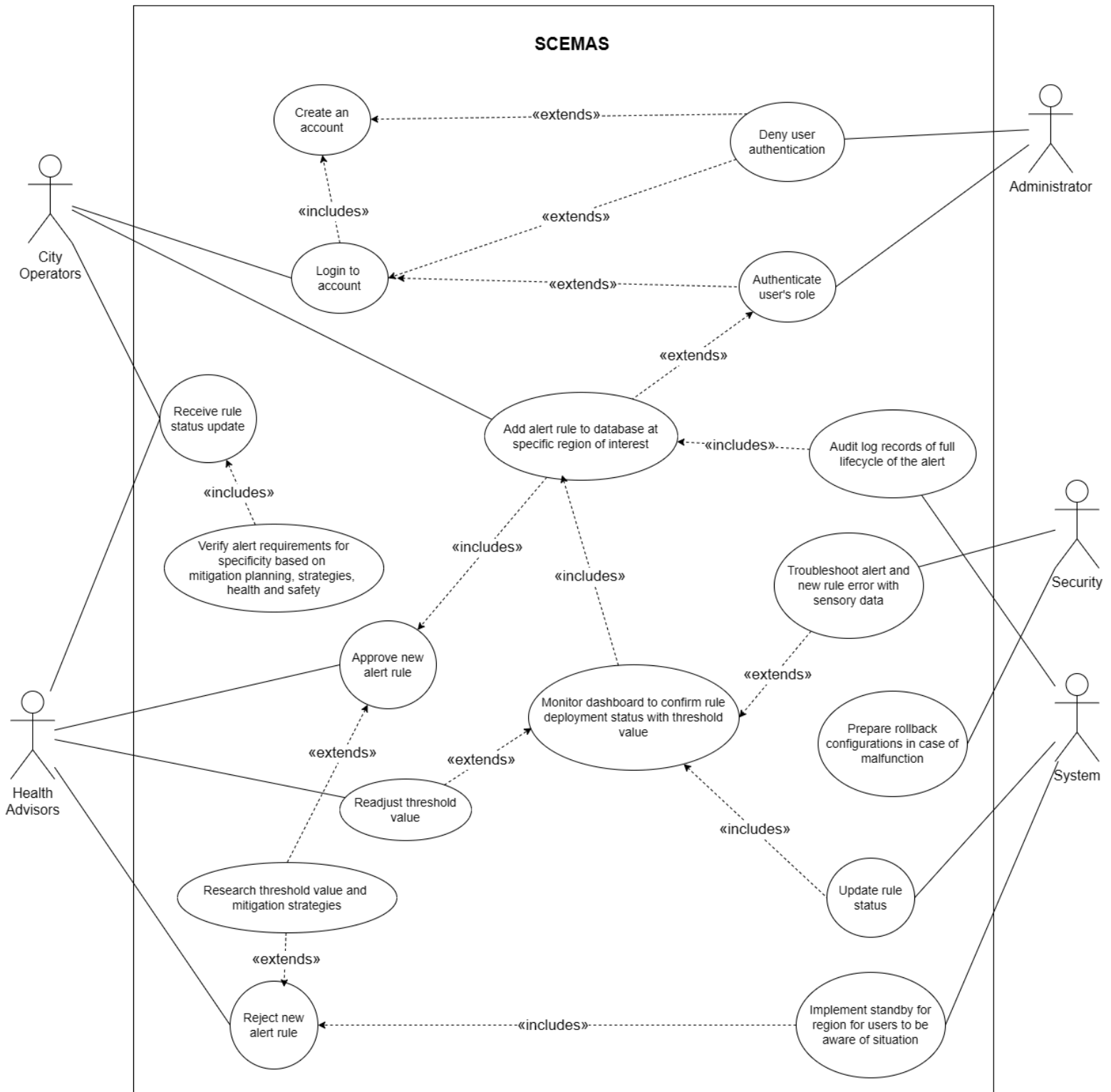


Figure 3. Use Case Diagram - Define new alert rule scenario

4 Highlights of Functional Requirements

The business events we will consider include:

- **BE1: Data Received From Sensor**
- **BE2: Define Geographic Region**
- **BE3: Define New Alert Rule**
- **BE4: Delete Alert Rule**
- **BE5: Delete Geographic Region**
- BE6: Create/Register Sensor (Nadeem)
- BE7: Delete Sensor (Muhammad)
- BE8: Read Live Data (Kaitlyn)
- BE9: Read Historic Data (Kurlan)
- BE10: Register Alert Callback (Aiden)
- **BE11: User Register**
- **BE12: User Login**
- **BE13: Request to Customize Dashboard**

The Viewpoints we will consider are:

- VP1: Legal/compliance
- VP2: Regular Citizen
- VP3: Third-party Developer (w/ API Access)
- VP4: City Operator
- VP5: Health Sector
- VP6: Security
- VP7: Finance

Business Events:

BE1. Data Received From Sensor #1:

Pre-Condition: API developer has subscribed to alerts pertinent to the datapoint, and IoT sensor is able to attest its identity to the server.

VP1. Legal/Compliance #1

1. Incoming sensor data must be authenticated.

VP2. Regular Citizen #2

Pre-Condition: Citizen is using app/service/etc that subscribes to alert notifications from our system.

Main Success Scenario

1. New data point received by server.
2. Data point checked against alert rules.
3. If data point outside defined threshold, trigger alert.
4. If alert triggered, send notification to all subscribed services.
5. Citizen receives notification about alert.

Secondary Scenario

- 3i: Data point inside alert thresholds.
- 3i.1: No alert triggered. Scenario ends.

VP3. Third-party Developer #3

Pre-Condition: API developer has subscribed to alerts pertinent to the datapoint.

Main Success Scenario

1. New data point received by server.
2. Data point checked against alert rules.
3. If an alert should be triggered, the third-party system is notified.
Otherwise, nothing happens.

VP4. City Operator #4**Main Scenario**

1. New data point received.
2. Aggregations of the relevant sensor region are updated.
3. Dashboard updated with new live data and updated aggregations.

VP5. Health Sector #5

VOID

VP6. Security #6

1. Data points should be received over an authenticated and encrypted connection.
 - Data cannot be read in flight by malicious actors, data coming from a sensor whose identity has been verified, and which cannot be spoofed by malicious actors.

VP7. Finance #7

VOID

Global Scenario:

Pre-Condition: IoT sensor is able to attest its identity to the server.

Main Scenario

1. New data point received by Ingest over network.

2. Ingest service stores raw data point, associated to its device's region.
3. Ingest service recalculates aggregated metrics.
4. Ingest service checks aggregated data stream, and the raw data point against alert rules.
5. Updated raw and aggregated data streams sent to web dashboard.

Secondary Scenario

- 4i. Alarm triggered.
 - 4i.1 Create an alarm with relevant metadata, store it in the database.
 - 4i.2 Lookup subscribers for who must be notified about this alert.
 - 4i.3 Transmit alert to relevant subscribers.
 - 4i.4 Citizens using subscribed services receive notification about alert.

BE2. Define Geographic Region #2

Pre-Condition: Operator is logged in to the system, operator has admin access.

VP1. Legal/compliance #1

1. The **geofence** should fall within city boundaries.
2. The **geofence** should be sufficiently large so as to anonymize aggregated sensor data.

VP2. Regular Citizen #2

VOID

VP3. Third-party Developer #3

VOID

VP4. City Operator #4

Main Success Scenario

1. Operator provides a set of coordinates defining a bounding box for the region.
2. Operator defines a name for the region.
3. The new region is stored in the database.

VP5. Health Sector #5

VOID

VP6. Security #6

1. Network traffic to create a region should be encrypted.

VP7: Finance #7

VOID

Global Scenario:

Pre-Condition: Operator is logged in to the system, operator has admin access, network traffic is encrypted.

Main Success Scenario

1. Operator provides a set of coordinates defining a bounding box for the region.
2. Operator defines a name for the region.
3. The new region is stored in the database.

Secondary Scenario

- 1i. **Geofence** falls outside city boundaries.
 - 1i.1 System returns an error to the user indicating coordinates that fall outside of the city boundary.
- 2i. **Geofenced** region is too small.
 - 2i.1 System returns an error to the user indicating the desired **geofence** is too small.

BE3. Define New Alert Rule #3

Pre-Condition A region is notified of an alert of potential hazards in the area and city and health administrators have access to aggregate data.

VP1. Legal/compliance #1**Main Success Scenario**

1. Legal team have to ensure that incoming data must be authenticated at the region.
2. Team must ensure that the threshold value must be authenticated by health advisors with regulatory standards.
3. Team observe information passed to system to ensure alert is based on the new rule.
4. Team should have testing/verification process between incoming telemetry data and comparison with the threshold value.
5. Team should ensure that the verification that the alert notification list complies with data sharing agreements.
6. Team should ensure audit actions taken by the system of the new alert has been created.

Secondary Scenario

- 1i. System fails to authenticate the incoming data.
 - 1i.1 If data is not authenticated, recheck sensor configuration setting.
 - 1i.2 If data is not authenticated, check network.
- 2i. If the alert has not been authenticated with advisors, the region will not have a threshold value of the alert and will be on standby (not guaranteed)

safety - be advised) of the specified region. The standby is removed until region obtain threshold by advisors.

- 5i. If alert notification is not sent to notified parties, a region will still be considered unsafe by threshold (not as an alert, but displayed with a message).

VP2. Regular Citizen #2

VOID

VP3. Third-party Developer #3

VOID

VP4. City Operator #4

Pre-Condition: City operator is logged in to the system and has admin access. Furthermore, with the addition that every sensor/MQTT client of a specific alert rule is sent in place at a region.

Main Success Scenario

1. The system authenticates the operator.
2. The system authorizes the operator, ensuring their account has the city operator role.
3. City operator navigates to alert configuration panel.
4. Operator selects “Edit Alert Rule” option.
5. Operator establish sensitivity alert parameters (metric, threshold, info).
6. Operator adds threshold value in place at that specific region .
7. Operator adds the new alert rule in the system database.
8. Operator observe the visual dashboard that it is updated continuously with raw data.
9. Operator continuously monitor vitals that new alert is in place at the region.
10. Operator observe that raw historical data trends to observe data to alert value (is it increasing or decreasing, or remain constant, is it following the new rule).

Secondary Scenario

- 5i. Threshold value does not have given parameters.
 - 5i.1 Operator have to contact with health administrators of research/literature review and discuss parametrics.
 - 5i.2 Use NaN parametric value until parameters are resolved.
- 10i. Threshold is observed to be happening quite frequently, meet with health administrators as well as configure sensory telemetry of any concern.
 - 10i.1 Ensure that health alert is aligned with mitigation strategies/warnings

about possibilities of hazards.

10i.2 Ensure that recorded data is aligned with sensory telemetry.

VP5. Health Sector #5

Pre-Condition: threshold has had extensive research done (these include metrics, symptoms, possible consequence, background, literature review, methodology and results).

Main Success Scenario

1. Health advisors review epidemiological standards before approving threshold values.
2. Advisors is authorized to ensure that the threshold aligns with public health guidelines.
3. Advisors ensure that have rule includes pollutant type, measurement unit, time window (referent window of epidemiological studies), and affected region.
4. Advisors gets approval of health authority from digitally approving threshold configuration.
5. Advisors follow that medical advisory board reviews and approves the new alert rule parameters.
6. Advisor add that the new alert rule will have dedicated severity level (advisory, warning, critical) which is mapped to health risk level.
7. Advisors add that the alert rule includes recommended public health actions.
8. Advisors communicate to place healthcare facility notification protocols are integrated, receive specialized notifications with clinical guidance.
9. Advisors add to place alert rule that indicates vulnerable population protection measures (elderly, children, respiratory patients) public health action.
10. Advisors ensure that the health sector receives confirmation of successful rule deployment.

Secondary Scenario

- 2i. Threshold exceeds medically accepted range.
 - 2i.1 System flags rule as “Pending Review”.
 - 2i.2 Health authority must re-approve before activation.
- 4i. Health authority approval token invalid.
 - 4i.1 Rule remains in draft state.
 - 4i.2 Other intensive research to validate threshold, need for more configuration and work before validation.
- 3i. Conflicting rule already exists for same pollutant and region.
 - 3i.1 System prompts merge/override decision.

3i.2 Health authority must confirm override.

VP6. Security #6

Main Success Scenario

1. Security initiates a secure session with multi-factor authentication.
2. Security role-based permissions for rule creation are validated for assigned roles for adding and managing threshold values.
3. System encrypts sensitive rule parameters during configuration.
4. Security has a rule definition that requests are encrypted
5. Security ensure that the input validation prevents injection or malformed data within the system.
6. Security checks that the rate limiting prevents abuse of rule creation endpoints.
7. Security follow that there is a mandatory integrity check to validate that there is no tampering during rule transmission.
8. Security implements the version control that maintains rule history for rollback capability.
9. Security ensures that the audit system log entry is created with timestamp and action details of every action within the system.

Secondary Scenario

- 3i. Suspicious activity detected (multiple rapid rule attempts).
 - 3i.1 Account temporarily locked.
 - 3i.2 Security team notified and manage role of the individual and/or role that the person(s) is in.
- 4i. Data integrity hash mismatch.
 - 4i.1 Rule creation rejected.
 - 4i.2 System triggers internal security warning and audits it for future analysis.

VP7. Finance #7

Main Success Scenario

1. Financers ensure that the new alert rule increases system processing load delegate cost planning.
2. Financers opens and have resource usage is monitored and alert evaluations scale efficiently.
3. Financers update operational costs and observe if it remain within budget within the scope of the project development.
4. Financers monitor if rule contributes to proactive mitigation saving into determining economic risk overtime.

Secondary Scenario

- 3i. Excessive alert rules cause performance degradation, need to improve network with the security team and compliance regulation.

Global Scenario:

Pre-Condition: Role-based access controls are met with all viewpoints. Security, health administrators and advisors, and city operators have full access to system setting.

Main Success Scenario

1. The health advisors have done extensive research to the alert threshold value to plan and find mitigation strategies if an alert was to be taken place.
2. The new alert rule is validated across legal, health, security, and financial domains.
3. The operator edits and stores the alert rule in the database at a specific region of interest.
4. The system audit log records the full lifecycle event of the alert being changed and/or edited by the operator.
5. The operator monitors the dashboard to confirm rule deployment status of health advisors to check if the data is consistent with the alert rule threshold value.
6. The operator and health administrators receive confirmation of the rule being in place.
7. Security will have rollback configuration prepared in case of malfunction were to occur in the system and handle error.
8. Advisor and operator will check if the rule becomes active in order to check if alerts have the right requirement for specificity based on mitigation planning, strategies, health and safety.

Secondary Scenario

- 2i. If new rule is not validated, an alert cannot be called (therefore a standby for a region is implemented for users to be cautious/aware of the situation).
- 5i. If issue with threshold and telemetry data, check in with security and health advisors.
 - 5i.1 For sensory data, stop flow of data until it is validated by the security team when configuration and troubleshooting.
 - 5i.2 For threshold value, alerts will not be sent until threshold is implemented and made changes.

BE4. Delete Alert Rule #4

Pre-Condition There are existing alert rules in **SCEMAS** alert engine, and the city operator is authorized and logged into the home page.

VP1. Legal #1

VOID

VP2. Citizen #2

VOID

VP3. Third-party Developer #3

4. If subscribed to the region the alert is deleted for, they are notified that an alert rule has been removed.

VP4. City Operator #4

Main Success Scenario

1. The operator selects the “View Alert Rules” option.
2. The operator enters the region they want to view defined alert rules for.
3. The operator selects the delete option next to one of the alert rules.
4. The system removes the alert rule from the alert engine for the selected region.

Secondary Scenario

- 4i. The system can delete can be searched and selected.
 - 4i.1 The operator can search for a specific environmental metric, which shows its corresponding alert rule.
 - 4i.2 The operator can select multiple alert rules to delete simultaneously.

VP5. Health Sector #5

3. Next to each alert, display a list of potential symptoms resulting from the alert’s threshold being exceeded.
4. Warn the operator of the potential consequences of removing the alert, including the potential health effects of not tracking its associated environmental metric (e.g. pollution).

VP6. Security #6

VOID

VP7. Finance #7

VOID

Global Scenario:

Pre-Condition: There are existing alert rules in **SCEMAS** alert engine, and the city operator is authorized and logged into the home page.

Main Success Scenario

1. The system redirects the operator to **SCEMAS** home page.
2. The operator selects the “View Alert Rules” option.
3. The operator enters the region they want to view defined alert rules for.
4. The operator selects the delete option next to one of the alert rules.
5. The operator is warned of the potential consequences of removing the alert.
6. The system removes the alert rule from the alert engine for the selected region.
7. If subscribed to the region the alert is deleted for, they are notified that an alert rule has been removed.

Secondary Scenario

- 4i. The system can delete can be searched and selected.
 - 4i.1 The operator can search for a specific environmental metric, which shows its corresponding alert rule.
 - 4i.2 The operator can select multiple alert rules to delete simultaneously.

BE5. Delete Geographic Region #5

Pre-Condition The region is no longer available to have alert rules sent by the legal/compliance team. City operators have access to deletion.

VP1. Legal/compliance #1**Main Success Scenario**

1. Legal team reviews region deletion request against regulatory requirements.
2. Team double checks system confirmation of minimum statutory data retention period has been satisfied.
3. Team ensure the integrity of the other regions have not been deleted.
4. Team checks the reason why the deletion has taken place with traceability.
5. Team then allows the process of deletion for the operator to the specific region.
6. Confirmation of compliant deletion is generated.
7. Team should ensure audit actions taken by the system of the deleted region.

Secondary Scenario

- 2i. Regulatory hold on region data.
 - 2i.1 Deletion blocked.
 - 2i.2 Legal team and city operator notified that deletion is not made.

3i. Team cannot check the traceability of the reason/concern of the removal of a specific region.

3i.1 Deletion blocked.

3.i.2) Legal team and city operator notified that deletion is not made.

VP2. Regular Citizen #2

VOID

VP3. Third-party Developer #3

VOID

VP4. City Operator #4

Pre-Condition: Operator is logged in to the system, operator has admin access.

Main Success Scenario

1. The system authenticates the operator.
2. The system authorizes the operator, ensuring their account has the city operator role.
3. City operator navigates to region configuration panel.
4. Operator selects the “Edit Region” option.
5. Operator selects the region where they want to remove the region for.
6. Operator selects the delete option next to one of the region available.
7. Operator confirms deletion via secondary confirmation prompt.
8. The system removes the region from region engine and database.
9. Operator receives confirmation of the deleted region.

Secondary Scenario

- 6i. Active critical alert exists.
 - 6i.1 Deletion blocked.
 - 6i.2 Operator instructed to resolve alert first.

VP5. Health Sector #5

VOID

VP6. Security #6

Main Success Scenario

1. Security initiates a secure session with multi-factor authentication.
2. Security role-based permissions for rule creation are validated for assigned roles for removal of region (it is only done by city operator).
3. Security checks if a request is transmitted.
4. Security ensure that no alert is present when deletion has taken place.

5. Security ensure that confirmation is sent to the directed operator for region deletion.
6. Security ensures that the audit system log entry is created with timestamp and action details of every action within the system.

Secondary Scenarios

- 2i. Unauthorized attempt detected.
 - 1i.1 Access denied.
 - 1i.2 Security alert triggered.
- 3.i) Suspicious repeated deletion attempts.
 - 3i.1 Account temporarily locked.
 - 3i.2 Security team notified and confirms the city operator and the decision made with legal team.

VP7. Finance #7

VOID

Global Scenario:

Pre-Condition: Role-based access controls are met with all viewpoints. Security, health administrators and advisors, and city operators have full access to system setting.

Main Success Scenario

1. The system authenticates the operator.
2. The system authorizes the operator, ensuring their account has the city operator role.
3. City operator navigates to region configuration panel.
4. Operator selects the “Edit Region” option.
5. Operator selects the region where they want to remove the region for.
6. Operator selects the delete option next to one of the region available.
7. Operator confirms deletion via secondary confirmation prompt.
8. The system removes the region from region engine and database.
9. Operator receives confirmation of the deleted region.
10. Security checks if a request is transmitted.
11. Security ensure that no alert is present when deletion has taken place.
12. Security ensure that confirmation is sent to the directed operator for region deletion.
13. Legal team checks the reason why the deletion has taken place with traceability.
14. Security ensures that the audit system log entry is created with timestamp and action details of every action within the system.

Secondary Scenarios

11i. Active critical alert exists.

11i.1 Deletion blocked.

11i.2 Operator instructed to resolve alert first.

13i Team cannot check the traceability of the reason/concern of the removal of a specific region.

13i.1 Deletion blocked.

13i.2 Legal team and city operator notified that deletion is not made.

BE6. Create/Register Sensor #6

Pre-Condition: there is a physical sensor at the correct location with an IP Address mapped to it. The city operator is logged in and has admin access.

VP1. Legal/compliance #1

1. The mapped Sensor's location, type, and IP are logged for auditing purposes and for legal reasons.

VP2. Regular Citizen #2

VOID

VP3. Third-party Developer #3

VOID

VP4. City Operator #4

Main Success Scenario

1. Operator goes to add sensor view.
2. Operator inputs relevant data such as location, IP, and type of sensor.
3. Operator adds sensor to relevant group(s) of sensor data.

VP5. Health Sector #5

VOID

VP6. Security #6

1. Verify Sensor won't be physically tampered with - must be safe from weather and out of the way of citizens.
2. Verify that IP address of Sensor is valid.
3. Verify that the sensor will be safe from cybersecurity attacks.

VP7. Finance #7

Main success:

1. Device will be added to network's list of registered sensors
2. Any operating costs from sending signals will be added to total sum of network costs.

Global Scenario:

Pre-Condition: there is a physical sensor at the correct location with an IP Address mapped to it. The city operator is logged in and has admin access.

1. Verify sensor won't be physically tampered with (weather and people).
2. Verify IP Address of sensor.
3. Verify sensor cybersecurity and safety from malicious attacks.
4. Operator goes to add sensor view.
5. Operator inputs relevant data such as location, IP, and type of sensor.
6. Operator adds sensor to relevant group(s) of sensor data.
7. Sensor adds System/operator IP as main address to send data to.

BE7. Delete Sensor #7

Pre-Condition: Operator is logged in, operator has admin access, operator's connection is encrypted, and a sensor exists and is registered to the network.

VP1. Legal/compliance #1

1. Any data received from a deleted device should be discarded.

VP2. Regular Citizen #2

VOID

VP3. Third-party Developer #3

VOID

VP4. City Operator #4

Main Success Scenario

1. Operator requests deletion of device.
2. Server removes device entry from database.

VP5. Health Sector #5

1. A deletion should be double checked by the user if it would delete the last device operating in a region.

VP6. Security #6

1. Deletion request should be encrypted over the network.

VP7. Finance #7:

1. Deleted devices should be disconnected to avoid parasitic network costs.
2. Hardware associated with deleted devices should be recovered.

Global Scenario:

Pre-Condition: Operator is logged in, operator has admin access, operator's connection is encrypted, and a sensor exists and is registered to the network.

Main Success Scenario

1. Operator requests deletion of device.
2. Server removes device entry from database.
3. Server sends a message instructing device to disconnect itself.
4. Future connection attempts by the deleted device will be rejected.

Secondary Scenarios

- 1i. Device is the last sensor in a given geographic region.
 - 1i.1 A warning is sent to the operator, and confirmation is requested before deletion is processed.
- 2i. Data received by device before disconnection.
 - 2i.1 Data ingress service discards any data by a device that has been deleted, but hasn't disconnected from the broker yet.

BE8. Read Live Data #8

Pre-Condition There are existing sensors set-up from the administrative end of the **SCEMAS** system, providing live data from the desired geographic location. Additionally, the city operator is logged into the **SCEMAS** home page.

VP1. Legal

VOID

VP2. Citizen

VOID

VP3. Third-party Developer

VOID

VP4. City Operator

Main Success Scenario

1. The system redirects the operator to **SCEMAS** home page.
2. The operator selects "View Live Data" option.
3. The operator enters a desired location to view live data for.

4. The System verifies live data is available for the desired region.
5. The system displays live data for the region.

Secondary Scenario

- 4i. Live data is unavailable for the desired region.
 - 4i.1 Live data is unavailable for the desired region.
 - 4i.2 Display a warning message to the operator that no data is available currently for the selected region.
- 5i. Operator selects “View Sensors” to see a geographical map of sensor locations in the region.
- 5ii. Operator selects “View System Health”, providing an overview of sensor health in the region.
- 5iii. Operator selects “View Historical Data”, providing the historical trends for different environmental metrics in the region.
- 5iv. Operator selects “View Active Alerts”, providing a list of currently active alerts in the region.

VP5. Health Sector

5. Display list of common health effects that result from active alerts.

VP6. Security

VOID

VP7. Finance

5. Display cost of replacing failing sensors in the region. (Out of Scope).

Global Scenario:

Pre-condition: There are existing sensors set-up from the administrative end of the **SCEMAS** system, providing historic data from the desired geographic location. Additionally, the city operator is logged into the **SCEMAS** home page.

Main Success Scenario

1. The operator selects “View Live Data” option.
2. The operator enters a desired location to view live data for.
3. System verifies live data is available for the desired region.
4. The system displays live data for the region.

Secondary Scenario

- 4i. Live data is unavailable for the desired region.
 - 4i.1 Live data is unavailable for the desired region.
 - 4i.2 Display a warning message to the operator that no data is available currently for the selected region.
- 5i. Operator selects “View Sensors” to see a geographical map of sensor

locations in the region.

5ii. Operator selects “View System Health”, providing an overview of sensor health in the region.

5ii.1 Operator selects “View System Health”, providing an overview of sensor health in the region.

5ii.2 Display cost of replacing failing sensors in the region.

5iii. Operator selects “View Historical Data”, providing the historical trends for different environmental metrics in the region.

5iv. Operator selects “View Active Alerts”, providing a list of currently active alerts in the region.

5iv.1 Operator selects “View Active Alerts”, providing a list of currently active alerts in the region.

5iv.2 Display common health effects that result from the active alerts.

BE9. Read Historical Data #9

Pre-Condition There are existing sensors set-up from the administrative end of the **SCEMAS** system, providing historic data from the desired geographic location. The user is also logged into the **SCEMAS** home page.

VP1. Legal #1

VOID

VP2. Citizen #2

Main Success Scenario

1. The user selects the “View Environmental Data” option.
2. The user enters a desired location to view historic data for.
3. The system verifies historic data is available for the desired region.
4. The user selects a time period (including a start and end date) to view aggregated data for.
5. The user selects an environmental metric to view data for.
6. The system displays the aggregated data to the user.

Secondary Scenario

- 3i. Historic data is unavailable for the desired region.
 - 3i.1 Historic data is unavailable for the desired region.
 - 3i.2 Display a warning message to the user that no data is available currently for the selected region.
- 4i. Historic data is unavailable for the specified time period.
 - 4i.1 Historic data is unavailable for the specified time period.
 - 4i.2 Display a warning message to the user that no data is available between the selected dates.

- 4i.3 The system will display the oldest date data is available from.
- 5i. Historic data is unavailable for the specified environmental metric.
 - 5i.1 Historic data is unavailable for the specified environmental metric.
 - 5i.2 Display a warning message to the user that no data is available for the selected environmental metric.
 - 5i.3 System will display alternative metrics the user may view for that region.

VP3. Third-party Developer #3

- 1. Third-party developers makes an API request with the desired region, start and end date, and environmental metric they want data for.
- 3.ii) API request returns an error if historic data is unavailable for the requested region.
- 4.ii) API request returns an error if historic data is unavailable within the selected date range.
- 5.ii) API request returns an error if historic data is unavailable for the specified environmental metric.

VP4. City Operator #4

VOID

VP5. Health Sector #5

VOID

VP6. Security #6

- 2.i) System temporarily disables region searching if the user makes too many requests in a short period of time to prevent API abuse.
- 6.i) Data must be aggregated and associated with general city zones, not precise locations.

VP7. Finance #7

N/A

Global Scenario:

Pre-condition: There are existing sensors set-up from the administrative end of the SCEMAS system, providing historic data from the desired geographic location. The user is also logged into the SCEMAS home page.

Main Success Scenario

- 1. The user selects the “View Environmental Data” option.

2. The user enters a desired location to view historic data for.
3. The system verifies historic data is available for the desired region.
4. The user selects a time period (including a start and end date) to view aggregated data for.
5. The user selects an environmental metric to view data for.
6. The system displays the aggregated data to the user.

Secondary Scenario

- 2i. System temporarily disables region searching if the user makes too many requests in a short period of time to prevent API abuse.
- 3i. Historic data is unavailable for the desired region.
 - 3i.1 Historic data is unavailable for the desired region.
 - 3i.2 Display a warning message to the user that no data is available currently for the selected region.
- 3ii. API request returns an error if historic data is unavailable for the requested region.
- 4i. Historic data is unavailable for the specified time period.
 - 4i.1 Historic data is unavailable for the specified time period.
 - 4i.2 Display a warning message to the user that no data is available between the selected dates.
 - 4i.3 System will display the oldest date data is available from.
- 4ii. API request returns an error if historic data is unavailable within the selected date range.
- 5i. Historic data is unavailable for the specified environmental metric.
 - 5i.1 Historic data is unavailable for the specified environmental metric.
 - 5i.2 Display a warning message to the user that no data is available for the selected environmental metric.
 - 5i.3 System will display alternative metrics the user may view for that region.
- 5ii. API request returns an error if historic data is unavailable for the specified environmental metric.

BE10. Register Alert Callback #10

Pre-Condition: User has created an account with SMS and/or email linked

VP1. Legal/compliance #1
VOID

VP2. Regular Citizen #2

Main Success Scenario

1. Citizen enters on app or website.
2. Citizen chooses sensor/aggregate sensor data they want alerts for.
3. Citizen inputs way to receive alerts (SMS or email).
4. Citizen verifies way of communication (verify email/send phone code).
5. Citizen confirms they want to receive alerts.

VP3. Third-party Developer #3

1. Third party developer connects to system API.
2. Developer sends data about where the sensor data should be sent.
3. Developer sends details about sensor/aggregate sensor data they want alerts for.

VP4. City Operator #4

1. System adds citizen/API call to list of addresses to be alerted for a sensor/aggregate sensor data.
2. System sends sensor data when there is relevant sensor data.

VP5. Health Sector #5

1. Alerts are sent out if there is a health risk for a certain area.

VP6. Security #6

VOID

VP7. Finance #7

1. Costs from sensor alerts are tallied up with total network costs

Global Scenario:

Precondition: User has created an account with SMS and/or email linked

1. Citizen enters on app or website.
2. Citizen chooses sensor/aggregate sensor data they want alerts for.
3. Citizen inputs way to receive alerts (SMS or email).
4. Citizen verifies way of communication (verify email/send phone code).
5. Citizen confirms they want to receive alerts.
6. System adds citizen to list of addresses to be alerted for a sensor/aggregate sensor data.
7. System sends sensor data when there is relevant sensor data.
8. Relevant charges applied for each alert sent.

Secondary Scenario

- 1i. Third party developer connects to system API.
- 2i. Developer sends data about where the sensor data should be sent.
- 3i. Developer sends details about sensor/aggregate sensor data they want alerts for.
- 4i. System adds API call to list of addresses to be alerted for a sensor/aggregate sensor data.
- 5i. System sends sensor data when there is relevant sensor data.
- 6i. Relevant charges applied for each alert sent.

BE11. User Register #11

Pre-Condition User does not have an existing account.

VP1. Legal/compliance #1

VOID

VP2. Regular Citizen #2

Main Success Scenario

1. Citizen opens registration page.
2. System asks for personal details.
3. Citizen enters required information.
4. The system authenticates the citizen.
5. System logs registration and sends to database.

Secondary Scenario

- 2i. Citizen enters wrong information.
 - 2i.1 System cannot accept input.
 - 2i.2 Citizen Register failed.
 - 2i.3 System displays error message.

VP3. Third-party Developer #3

Main Success Scenario

1. Developer opens registration page.
2. System asks for personal details.
3. Developer enters required information.
4. Developer requests for API access.
5. The system authenticates the developer with a standard account.
6. Admin reviews request and accepts.
7. The system grants API access to the developer's account.
8. System displays success message.

Secondary Scenario

- 5i. Developer enters wrong information.

- 5i.1 System cannot accept input.
- 5i.2 Developer Register failed.
- 5i.3 System displays error message.
- 6i. Admin rejects API request.
 - 6i.1 Developer stays on standard account.
 - 6i.2 System displays rejection message.

VP4. City Operator #4

Main Success Scenario

1. Operator opens registration page.
2. System asks for personal details.
3. Operator enters required details.
4. Operator enters organization details.
5. System authenticates operator with standard account.
6. Admin verifies organization and approves.
7. System grants organization access to operator's account.
8. System displays success message.

Secondary Scenario

- 5i. Operator enters wrong information.
 - 5i.1 System cannot accept input.
 - 5i.2 Operator Register failed.
 - 5i.3 System displays error message.
- 6i. Admin rejects organization.
 - 6i.1 Operator stays on standard account.
 - 6i.2 System displays rejection message.

VP5. Health Sector #5

VOID

VP6. Security #6

- 7. Assign role-based access at account creation.

VP7. Finance #7

VOID

Global Scenario:

Pre condition: User does not have an existing account.

Main Success Scenario

1. User opens registration page.
2. System asks for personal details.

3. User enters required information.
4. System authenticates user with standard account.

Secondary Scenario

- 3i. User chooses to enter role-specific details (request API or organization).
 - 3i.1 Admin verifies role specific requests and accepts.
 - 3i.2 System grants specific role-based access.
 - 3i.3 System displays success message.
- 3ii. Admin rejects request.
 - 3ii.1 User stays on standard account.
 - 3ii.2 System displays rejection message.
- 4i. User entered incorrect information.
 - 4i.1 System detects wrong information.
 - 4i.2 User registration failed.
 - 4i.3 System displays error message.

BE12. User Login #12

Pre condition User does have an existing account.

VP1. Legal/compliance #1

VOID

VP2. Regular Citizen #2

Main Success Scenario

1. Citizen navigates to login page.
2. System displays login inputs.
3. Citizen enters login credentials.
4. System authenticates user.
5. Citizen is redirected to the main dashboard.

Secondary Scenario

- 4i. Incorrect user information.
 - 4i.1 System detects incorrect user information.
 - 4i.2 User login failed.
 - 4i.3 System displays error message.

VP3. Third-party Developer #3

Main Success Scenario

1. Developer navigates to login page.
2. System displays login inputs.
3. The developer enters login credentials.
4. System authenticates user.

5. The developer is redirected to the developer dashboard.

Secondary Scenario

- 4i. Incorrect user information.
 - 4i.1 System detects incorrect user information.
 - 4i.2 User login failed.
 - 4i.3 System displays error message.
- 4ii. Access denied.
 - 4ii.1 System does not detect account to be developer.
 - 4ii.2 Developer login failed.
 - 4ii.3 System displays error message.

VP4. City Operator #4

Main Success Scenario

1. Operator navigates to login page.
2. System displays login inputs.
3. Operator enters login credentials.
4. System authenticates user.
5. The operator is redirected to the organization dashboard.

Secondary Scenario

- 4i. Incorrect user information.
 - 4i.1 System detects incorrect user information.
 - 4i.2 User login failed.
 - 4i.3 System displays error message.
- 4ii. Access denied.
 - 4ii.1 System does not detect account to be operator.
 - 4ii.2 Operator login failed.
 - 4ii.3 System displays error message.
- 4iii. Incorrect organization.
 - 4iii.1 System detects correct role of operator.
 - 4iii.2 System detect wrong organization linked to account.
 - 4iii.3 Operator login failed.
 - 4iii.4 System displays error message.

VP5. Health Sector #5

VOID

VP6. Security #6

4. Verifying role-based access when logged in.
 - 4i. The system prompts the user with “Forgot Password?” to allow them to reset their account password through their email. (Out of Scope).

VP7. Finance #7

VOID

Global Scenario:

Pre condition User does have an existing account.

Main Success Scenario

1. User navigates to login page.
2. System displays login inputs.
3. User enters login credentials.
4. User enters specific role information.
5. System verifies role based access.
6. System authenticates user.
7. The user is redirected to the appropriate dashboard.

Secondary Scenario

- 6i. Incorrect user information.
 - 6i.1 System detects incorrect user information.
 - 6i.2 User login failed.
 - 6i.3 System displays error message.
- 5ii. Access denied.
 - 5ii.1 System does not detect the role linked to the account to have access.
 - 5ii.2 User login failed.
 - 5ii.3 System displays error message.
- 5iii. Incorrect organization.
 - 5iii.1 System detects role of operator.
 - 5iii.2 System detect wrong organization linked to account.
 - 5iii.3 Operator login failed.
 - 5iii.4 System displays error message.

BE13. Request to Customize Dashboard #13

Pre-Condition User is logged in.

VP1. Legal/compliance #1

VOID

VP2. Regular Citizen #2

~~**Main Success Scenario**~~

- ~~1. Citizen navigates to dashboard settings.~~
- ~~2. Citizen requests customization.~~
- ~~3. System verifies role.~~

- ~~4. System gives limited customization access.~~
- ~~5. Citizen customizes dashboard and saves changes.~~
- ~~6. System approves changes and saves them to citizen's account.~~

~~— Secondary Scenario~~

- ~~— 5i. Dashboard changes not accepted.~~
- ~~— 5i.1 System denies changes made.~~
- ~~— 5i.2 System displays error message.~~
- ~~— 5i.3 Head back to BE13.VP2.1.~~

VOID

VP3. Third-party Developer #3

~~— Main Success Scenario~~

- ~~1. Developer navigates to dashboard settings.~~
- ~~2. Developer requests customization.~~
- ~~3. System verifies role.~~
- ~~4. System gives customization access with features for developers only.~~
- ~~5. Developer customizes dashboard and saves changes.~~
- ~~6. System approves changes and saves them to developer's account.~~

~~— Secondary Scenario~~

- ~~— 3i. Role access denied.~~
- ~~— 3i.1 System detects wrong role for developer access.~~
- ~~— 3i.2 System rejects customization and displays error message.~~
- ~~— 3i.3 Redirect to BE13.VP3.1.~~
- ~~— 5i. Dashboard changes not accepted.~~
- ~~— 5i.1 System denies changes made.~~
- ~~— 5i.2 System displays error message.~~
- ~~— 5i.3 Head back to BE13.VP3.1.~~

VOID

VP4. City Operator #4

Main Success Scenario

1. Operator navigates to dashboard settings.
2. Operator requests customization.
3. System verifies role.
4. System gives customization access with features for operators only.
5. Operator customizes dashboard and saves changes.
6. System approves changes and saves them.

Secondary Scenario

- 3i. Role access denied.
 - 3i.1 System detects wrong role for operator access.
 - 3i.2 System rejects customization and displays error message.
 - 3i.3 Redirect to BE13.VP4.1.
- 5i. Dashboard changes not accepted.
 - 5i.1 System denies changes made.
 - 5i.2 System displays error message.
 - 5i.3 Head back to BE13.VP4.1.

VP5. Health Sector #5

VOID

VP6. Security #6

3. System verifies role-based permissions and only gives access to correct data.

VP7. Finance #7

VOID

Global Scenario:

Pre condition: User is logged in.

Main Success Scenario

1. User navigates to dashboard settings
 - User requests customization.
2. System verifies user's role.
3. System gives customization access with correct permissions for user's role.
4. User customizes dashboard and saves changes.
5. System approves changes and saves them to user's account.

Secondary Scenario

- 3i. Role access denied
 - 3i.1 System detects wrong role for operator access.
 - 3i.2 System rejects customization and displays error message.
 - 3i.3 Redirect to BE13.VP4.1.
- 5i. Dashboard changes not accepted.
 - 5i.1 System denies changes made.
 - 5i.2 System displays error message.
 - 5i.3 Head back to BE13.VP4.1.

5 Non-Functional Requirements

5.1 Look and Feel Requirements

5.1.1 Appearance Requirements

LF-A1. Administrators must be able to define theme colours and fonts to be used throughout the application

Rationale: SCEMAS will be deployed by local governments for their jurisdictions, and city officials will want consistent branding across their internal systems.

LF-A2. The system shall have a user-friendly design

Rationale: SCEMAS will have a clean and simple design that will facilitate the use by any new user. It will also allow more efficient operations, since it would be quick to understand the layout and become faster at navigating through the system.

LF-A3. The system shall have colour contrast that conforms to WCAG

Rationale: SCEMAS will verify color contrasts for text, buttons, backgrounds and other elements of the app. This is to ensure that the system is accessible to everyone.

5.1.2 Style Requirements

LF-S1. The system must fit to any commonly used phone or monitor screen size.

Rationale: SCEMAS will be used by lots of different people with lots of different devices that can be accessible to everyone.

LF-S2. The system cannot show more than three distinct time series visualizations of data streams at a time.

Rationale: SCEMAS will be allowing data to be displayed to individuals using visual graphs to make judgments about the region without much background knowledge of parametric values. The graphs are limited so that it can be digestible and does not cluster the interface.

LF-S3. The system must navigate through scroll and should flow uniformly

Rationale: SCEMAS will allow users to easily navigate the app with an easy to read interface and move through the app quickly without wasting time and effort looking to subscribe to an alert.

5.2 Usability and Humanity Requirements

5.2.1 Ease of Use Requirements

UH-EOU1. Users must be able to report bugs/anomalies for the **SCEMAS** interface or for an individual sensor.

Rationale: Users should be able to report bugs so that the developers can address any issues.

5.2.2 Personalization and Internationalization Requirements

UH-PI1. The system must allow users to select their preferred language.

Rationale: **SCEMAS** should be deployable in a variety of different regions. International deployments will require internationalization features, including different UI languages.

UH-PI2. Users should be able to select common disability aiding settings. For **SCEMAS** this is defined as screen reader mode, high contrast mode, and color-blind mode

Rationale: All types of users will be using **SCEMAS**, they should all be able to easily use it and choose a better interface experience.

5.2.3 Learning Requirements

UH-L1. A trained city operator must be able to log into the dashboard and successfully acknowledge a critical environmental alert within 3 seconds of the alert appearing.

Rationale: Alerts could indicate emergency situations, an operator should not be confused while dealing with the system when they are in a high pressure scenario.

UH-L2. The public API must enable a third-party developer to successfully request and interpret environmental data within a two-hour timeframe.

Rationale: One of the main goals of **SCEMAS** is to encourage public awareness of city pollution levels, if someone is not able to properly use the API they will be dissuaded from using the application.

5.2.4 Understandability and Politeness Requirements

UH-UP1. All user-facing messages must be written in clear, concise, and non-technical language.

Rationale: Ensures that users with varying technical backgrounds can easily understand system messages, improving usability and reducing misinterpretation.

5.2.5 Accessibility Requirements

UH-A1. The system shall deliver alerts through a combination of auditory, haptic feedback, and visual notifications.

Rationale: Providing a combination of auditory, physical, and visual feedback ensures users with hearing loss, visual impairment, or hypoesthesia can still receive alerts. One or more of these alert notification methods can be optionally disabled.

5.3 Performance Requirements

5.3.1 Speed and Latency Requirements

PR-SL1. The response time of the read-only API used by third-party developers should be under 2s.

Rationale: Acceptable API response times are considered to be between 1 and 2 seconds [X] (<https://odown.com/blog/what-is-a-good-api-response-time/>). If the API response times are greater than 2s, users may become impatient and opt for alternative API services.

5.3.2 Safety-Critical Requirements

PR-SC1. The alert must contain an appropriate response and/or call for action that is delivered to the intended user of the regions that it is subscribed to.

Rationale: Providing a clear, actionable response within an environmental alert is essential to ensuring public safety and enabling timely mitigation of health risks. Research done in emergency communication has shown that alerts without explicit recommended actions lead to confusion, extreme panic, and reduced compliance [2].

5.3.3 Precision or Accuracy Requirements

PR-PA1. Sensor data collected by the platform must be within $\pm 10\%$ threshold of the true value.

Rationale: Inaccurate sensor data may result in safety critical alerts not triggering, compromising the health and safety of citizens in the affected area.

5.3.4 Reliability and Availability Requirements

PR-RA1. The uptime of the read-only API must exceed 99.99%.

Rationale: To ensure sensor data, system health, and key environmental metrics are available to city operators and the public, system downtime must be limited to a maximum of 1 minute and 30 seconds per day [3]. Increased downtime may result in a loss of trust in the platform, and missed alerts that may be safety critical [4].

5.3.5 Robustness or Fault-Tolerance Requirements

PR-RFT1. The system must be able to gracefully handle malformed sensor data

Rationale: Even with robust security measures in place to ensure that only authorized sensors can connect to the system, malformed data from sensors should not be able to crash or otherwise cause availability issues to the server.

5.3.6 Capacity Requirements

PR-C1. The system must be able to support at least 200 sensors in a **geofenced** region.

Rationale: Supporting several hundred sensors per region ensures the system can effectively aggregate sensor data by minimizing errors and inaccuracies in the measurement process.

PR-C2. The system must be able to support up to 10,000 users registered for alerts in their city.

Rationale: The average Canadian city has between 10,000 and 100,000 residents [5]. Assuming the high-end estimate of 100,000 residents in a city, and estimating that 10% of residents will register for alert callbacks in their region, up to 10,000 users must be supported.

5.3.7 Scalability or Extensibility Requirements

PR-SE1. The system cannot contain classes that exceed 1000 lines of code, or functions that exceed 300 lines of code.

Rationale: To encourage modularity and extensible code, class and function lengths should be limited [6]. Additional classes and functions can be created to support complex functionality, rather than creating large monolithic components that are difficult to scale and maintain.

5.3.8 Longevity Requirements

VOID

5.4 Operational and Environmental Requirements

5.4.1 Expected Physical Environment

OE-EPE1. Sensors should be able to withstand temperatures within the region of -25C to +65C

Rationale: These temperatures correspond roughly to the lowest winter temperature and the highest pavement temperature in Toronto ON over the past year (sensors will be out in the sun, they will heat up more than the air temperature), there is also a small safety margin added [7] [8]. Sensors should be able to operate in the full scale of climates in city deployments without issue.

OE-EPE2. Sensors should be able to withstand heavy precipitation, such as continuous snowfall up to 20cm without human intervention.

Rationale: Devices will be exposed to the elements, and should be resilient to typical snow and rainstorms without degradation of service. At the same time, it may not be practical to design devices to function without human intervention during the most extreme winter storms.

5.4.2 Requirements for Interfacing with Adjacent Systems

OE-IA1. System must be able to interface with an external MQTT broker

Rationale: MQTT must be used for interfacing with sensors, therefore our system must interface with an MQTT broker to subscribe to device topics.

5.4.3 Productization Requirements

OE-P1. System will be distributed as containerized applications for deployment to cloud platforms.

Rationale: **SCEMAS** is designed to be cloud-native, and modern cloud applications are most frequently deployed in containerized systems like Docker or Kubernetes.

5.4.4 Release Requirements

OE-R1. The app must be compatible with Android 5.0 or above.

Rationale: Supporting Android 5.0+ ensures compatibility with a wide range of devices still in circulation, increasing accessibility and maximizing the number of potential users.

OE-R2. The web dashboard must support all major modern browsers (Chrome, Firefox, Edge, Safari).

Rationale: Ensuring cross-browser compatibility allows city operators, health officials, and third-party stakeholders to access the system regardless of their preferred platform, reducing friction of access.

OE-R3. The public API must follow REST standards and be compatible with any HTTP-compliant client.

Rationale: RESTful APIs ensure interoperability across diverse systems, enabling third-party developers, signage systems, and institutions to integrate with **SCEMAS** for readability.

OE-R4. The system must support deployment on major cloud platforms (AWS, Azure, GCP) or on-premise Kubernetes cluster.

Rationale: Multi-platform deployment flexibility ensures that municipalities with different infrastructure constraints can adopt **SCEMAS** without vendor lock-in.

OE-R5. The IoT ingestion service must support MQTT 3.1.1 and MQTT 5.0.

Rationale: These versions are widely adopted in IoT ecosystems, ensuring compatibility with a broad range of sensors for environmental data.

OE-R6. The system must support TLS 1.2 or higher for all data-in-transit encryption.

Rationale: TLS 1.2+ is the current industry standard for secure communication, ensuring compliance with cybersecurity best practices and municipal IT policies.

OE-R7. The public-facing signage client must run on any modern browser or embedded OS supporting HTML5.

Rationale: Many digital signage systems use lightweight embedded browsers; HTML5 compatibility ensures the display client works across diverse hardware (laptops, phones, tablets, etc).

5.5 Maintainability and Support Requirements

5.5.1 Maintenance Requirements

MS-M1. The system must have bi-weekly hotfixes to address security concerns and high-impact bugs, when applicable.

Rationale: To maintain user and city operator trust in the platform, bi-weekly hotfixes to the software must be pushed. This requirement reduces future costs by minimizing the accumulation of technical debt and mitigating security breaches.

5.5.2 Supportability Requirements

MS-S1. The system must provide access to an active support email from the home page. Emails must be responded to within 1-2 business days.

Rationale: To ensure support is available to users, a dedicated support email must be displayed on the home page. Providing readily available support improves user trust in the system, and ensures issues identified with the platform can be addressed during the bi-weekly hotfixes.

5.5.3 Adaptability Requirements

VOID

5.6 Security Requirements

5.6.1 Access Requirements

SR-AC1. City Operators must be authenticated prior to accessing the dashboard

Rationale: Project description stipulates the web dashboard must be secure and role authenticated.

SR-AC2. API requests must be authenticated with a valid API key

Rationale: Project description stipulates that rate-limiting must be applied to mitigate DoS and unauthorized scraping. Requiring API keys allows precise per-application rate limiting, and will make unauthorized scraping impossible.

SR-AC3. All sensors must be authenticated when connecting to the MQTT broker.

Rationale: Project description stipulates that IoT devices must complete authentication prior to data transfer. Sensor authentication also ensures that data published over MQTT must originate from a trusted source.

5.6.2 Integrity Requirements

SR-INT1. All sensor data must originate from authenticated devices

Rationale: Data poisoning by unauthorized IoT devices could lead to false positive alarms, and would pollute aggregated metrics used by human operators, lessening the effectiveness of the platform.

SR-INT2. All incoming sensor data must be validated prior to storage.

Rationale: Project description stipulates that the system must validate incoming sensor streams. Data corruption in transit could also impact service effectiveness with false readings.

5.6.3 Privacy Requirements

SR-P1. All sensor data must be stored in anonymized, aggregated form.

Rationale: Project description stipulates that sensor data must be stored in aggregated form.

5.6.4 Audit Requirements

SR-AU1. The following actions must be logged for audit purposes, including metadata about the date and time of the operation, and the user who performed the operation.

- User account creation by an administrator
- User account deletion by an administrator
- User role change

- IoT device creation
- IoT device deletion
- IoT device modification
- Region creation
- Region deletion
- Alert rule creation
- Alert rule deletion
- Alert rule modification

Rationale: Project description stipulates audit logging. A malicious or compromised administrator account should not be able to take actions impacting the effectiveness of the platform without a paper trail.

SR-AU2. All accesses made by an API key must be logged for rate limit and audit purposes.

Rationale: Project description requires rate limiting on the public-facing API, logging will facilitate this. Furthermore, detailed API key usage logs will indicate whether a previously authorized credential has been compromised.

SR-AU3. Audit logs entries must not be deleted

Rationale: It should not be possible for a malicious actor to take actions, and hide them by removing associated audit log entries.

5.6.5 Immunity Requirements

SR-IM1. System must not allow incoming external connections, except to web services

Rationale: Exposing additional ports (e.g. database) to the internet presents a higher attack surface that may be exploited to inject malicious code.

SR-IM2. System's web processes should run behind a **WAF**

Rationale: A **WAF** will ensure HTTP requests are sanitized prior to arriving at our service, reducing the potential malware attack surface even further.

5.7 Cultural and Political Requirements

5.7.1 Cultural Requirements

CP-C1. The system must support custom icons and images from the administrator dashboard to replace symbols that may be considered offensive in certain regions.

Rationale: Default symbols, icons, and images may be considered offensive in various cultures and regions. Administrators may replace symbols, icons, and images to match their region's cultural context, ensuring the app feels welcoming and inclusive across all regions.

5.7.2 Political Requirements

VOID

5.8 Legal Requirements

5.8.1 Compliance Requirements

LR-COMP1. The collection of personal information must be limited to only information necessary for the application's functionality.

Rationale: PIPEDA Principle 4 - Limiting Collection is necessary to comply with PIPEDA requirements [9].

LR-COMP2. The system must identify the purposes for which it is collecting personal information, and make it clear to users before collection occurs.

Rationale: PIPEDA Principle 2 - Identifying Purposes is necessary to comply with PIPEDA requirements [9].

LR-COMP3. The system must allow the user to update their personal information to ensure stored info is accurate, complete, and up-to-date.

Rationale: PIPEDA Principle 6 - Accuracy is necessary to comply with PIPEDA requirements [9].

5.8.2 Standards Requirements

LR-STD1. The system must comply with all applicable privacy and confidentiality standards, including municipal data-protection policies and ISO/IEC 29100.

Rationale: This is a standard for app quality in terms of privacy and security, ensures that environmental data is handled responsibly and that the system aligns with recognized privacy principles [10].

6 Innovative Feature

Our team chose three innovative features to move forward with. We want to implement a customizable dashboard, predictions based on historic sensor and region data, and a hotspot identification system (for example, identifying areas with the worst air or areas that are the loudest). The customizable dashboard will provide a better user experience for city organizers

and citizens. The predictions based on historic data and the hotspot identification features will help gather more meaningful/better quality data and allow us to draw more conclusions about sensor data, this increases the perceived benefit of the application. This is a list of all the other creative features we came up with:

- Recommendation of Coldspots (opposite of Hotspots)
- Recommendation of alert resolution
- Baymax, send robot to resolve alerts
- Forecasting for pedestrians/cyclists (Based on air quality, precipitation, etc)

A Division of Labour

Include a Division of Labour sheet which indicates the contributions of each team member. This sheet must be signed by all team members.

Aiden Sanvido

- Section 2.2
- State diagram
- Read aggregated data, live data, and delete alert rules business events
- Brainstormed sensor data prediction for creative idea
- Reformatted headings, subheadings, and general document structure
- Contributed to style, learning, accessibility, speed and latency, precision and accuracy, reliability and availability, scalability, maintenance and supportability, cultural, and compliance non-functional requirements
- Completed IEE citations
- Reviewed and revised document as a group
- Brainstormed viewpoints and business events as a group



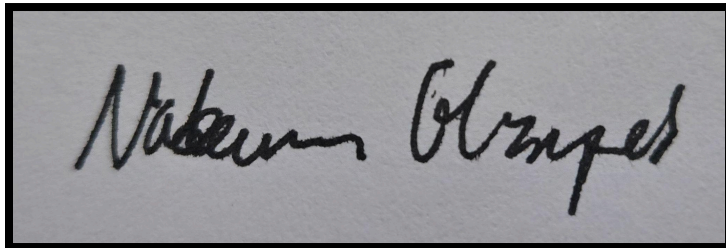
Muhammad Ullah

- Section 1.3
- Section 2.1
- System diagram
- Define alert rule and delete region business events
- Brainstormed creative idea of health safety
- Formatted business events and double checks
- Contributed to safety, release, and standards non-functional requirements
- Added IEEE citation
- Brainstormed viewpoints and business events as a group

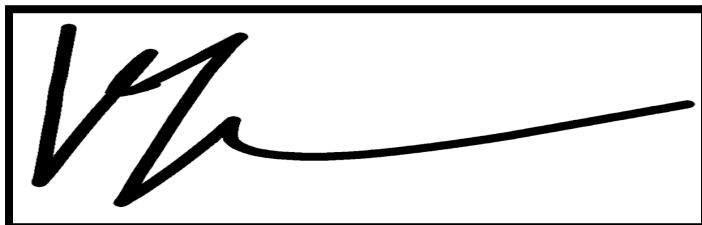


Nadeem Elsayed

- Section 1.1 - Introduction
- Section 1.2 - Scope
- Section 1.5 - Overview
- Section 6 - Innovative feature(s)
- Business Event 8 with viewpoints - Creating a sensor
- Business Event 9 with viewpoints - Registering Alert Callback
- Edited Section 5.4.1 - Expected Physical Environment
- Reviewed non-functional requirements as a team
- Brainstormed creative features with the team
- Reviewed and edited documents as a team

A black and white photograph of a handwritten signature, "Nadeem Elsayed", in cursive script. The signature is written in dark ink on a light-colored, slightly textured background. The entire signature is enclosed within a thin black rectangular border.**Kaitlyn Kenwell**

- Section 2.3
- Section 2.4
- Section 2.5
- Section 2.6
- Business Event 1 with viewpoints - Data Received from Sensor
- Business Event 2 with viewpoints - Define Geographic Region
- Business Event 7 with viewpoints - Delete sensor
- Brainstormed creative feature with team
- Contributed to Appearance, Robustness, Environmental, Interfacing, Productization, Release, Access, Integrity, Privacy, Audit, Immunity non-functional requirements.
- Brainstormed viewpoints and business events as a group

A black and white photograph of a handwritten signature, "Kaitlyn Kenwell", in a stylized, cursive script. The signature is written in dark ink on a light-colored background. The entire signature is enclosed within a thin black rectangular border.

Kurlan Beeharry

- Section 3 - Use Case Diagram
- Section 4.11 - User Login
- Section 4.12 - User Register
- Section 4.13 - Request Customization of Dashboard
- Section 5.1.1
- Contributed to Section 5.1.2
- Brainstormed creative features
- Brainstormed viewpoints and business events as a group
- Reviewed and revised document as a group
- Final check (Added page number and fixed some formatting) and submission

